U209

Composing three reductions for a 209-state consistency-machine candidate

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Abstract

The three supplied U211 proposals are distinct machines with a common 212-state ancestor. Their final deletions remove different native states: 180, 181, and 218. They cannot simply be applied together because they require different source associations and helper-state roles. This report constructs a common source layout that supports all three deletions, yielding an explicit binary, one-tape machine with **209 working states**, 418 transition rows, and a separate halt label. The new construction uses closed source reassociation, transports a 78-primitive continuation through a larger source region, and relocates one selector from native 129 to 217. The simultaneous final fold has an exhaustive ten-case, whole-tape proof. Every old step is replaced by one, three, or five new steps.

The released machine was rebuilt independently, its complete source control flow was checked, its exact 48,140-step blank initialization was replayed, and all 5,084 primitive contracts were derived over arbitrary natural registers and finite spare lists. All 418 literal rows are covered. Composing these results with the retained arithmetic calibration gives the conventional PA-formalizable conclusion $\mathrm{PA} \vdash \neg\,\mathrm{Halt}(U209, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})$. The archive supplies exact tables, source, proof arguments, reproducible checkers, and the inherited calibration. It supplies no newly kernel-checked PA, Lean, Rocq, or MM0 endpoint and makes no optimality claim.

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1 Result, machine convention, and proof status

The main result is a concrete reduction from each supplied 211-state upper bound to a common **209-state candidate**. The reduction is new relative to the inspected packets: the individual three deletion ideas are inherited, while their compatible source layout, selector relocation, and simultaneous composition are established here. This is not a claim of priority over uninspected work.

The model has one tape indexed by $\mathbb{Z}$, alphabet $\{0, 1\}$, blank symbol 0, and moves L or R at every transition. The dense table has working states $0, \ldots, 208$, initial state 0, and the separate halt label 209. The head starts at coordinate 0 on the all-zero tape. Initialization is included in the 209 states. There is no auxiliary tape, preloaded program, stationary move, or uncounted initialization routine.

The complete table is in Appendix A and `machine/u209.tm`. Native labels, retained for comparison with the input reports, appear in `machine/u209.native.tm`; native halt is 254. The exact bijection between the two tables is delivered and checked. Every working state is graph-reachable from the start. Graph reachability is a census check, not a claim that every state is actually visited on the unknown blank computation.

The dense table SHA-256 is

`3a69998d818b6e637e33dc0a2412538e49179a501496d5c252c7f68ff2af78be.`

The native table SHA-256 is

`8e058789f2f11a448db7212bf2e32317fbd5027c6043cfd0e7791532bc3256cb.`

Layer	What is established and how
Literal machine	Complete 209-state table; exact deterministic rebuild; dense/native isomorphism; target closure and graph census.
New local theorem	Ten two-cell identities plus 414 ordinary-row identities; halting preservation and reflection for the final simultaneous fold.
Source and runtime	Same 3,909 ordered primitive effects and all ordinal successors; complete final-table symbolic induction over unbounded inputs.
Blank start	Exact finite-support tape equality after 48,140 steps, at the same calibrated interior source occurrence.
Arithmetic endpoint	Composition with the supplied numerical-calculus, startup protection, fair-search, and U/PA interpretation proofs.
Formal certificate	No newly checked end-to-end PA, Lean, Rocq, or MM0 proof object is included. Python replay is not such a certificate.

Table 1: The verification layers are kept separate throughout the report.

The externally formalized K215 result concerns a different literal machine and ZFC [5, 6]. It is useful architectural background; its theorem does not automatically certify any U211 or U209 table. The present arithmetic conclusion uses the U-specific calibration reviewed in Section 6. No unconditional assertion of Con(PA) or of blank-tape nonhalting is made.

2 Comparison of the three supplied 211-state proposals

2.1 Pinned review scope

The repository snapshot is ProveIt commit

`da4ab9dd68ccfc47ac852118601eaf1fdae53795`.

At this snapshot `docs/incoming` contains exactly the three U211 archives and the unary-theory archive identified in the question. All four were downloaded, their Git blob identities verified, and their reports, builders, tables, and relevant proof material inspected. Their original bytes are retained under `reference/`. The source manifest records commit-pinned URLs, byte counts, Git blob hashes, and SHA-256 hashes.

The abbreviations below mean the files with suffix `_2026-10-28.zip`: Contextual is `U211_ConPA_Contextual`, Research is `U211_ConPA_Research`, and Rotated is `U211_ConPA_Rotated` [1, 2, 3]. The Research archive's inner directory is named `U211_ConPA_Decoder_Reassociation_2026-10-28`.

Proposal	Final deletion	Construction and distinctive contribution
Contextual	$180 \mapsto 163$	Six closed source rotations, an arithmetic entry repair at 171/0, and a predecessor-bit selector through 211. Retains the complete nested calibration ancestry.
Research	$181 \mapsto 220$	Nine rotations of a larger closed region; uses the predecessor bit through helper 129 to distinguish two selectors with a common one branch. Clear separation of rotation and fold scopes.
Rotated	$218 \mapsto 157$	Eleven rotations make every incoming edge of 218 leave a zero predecessor. Existing state 157 then implements both branches with a three-step detour.

Table 2: Three different last-state savings from the common U212 ancestor. Native labels are used.

The native SHA-256 prefixes are respectively `ac75e2fa934e`, `7edae566b4c2`, and `72234980bd60`. They are therefore not duplicate tables under the inherited labels. More substantively, their source addresses differ and they remove different states. All retain the same 3,909 primitive occurrences, 197 distinct source subtrees, and the same ordinal register program. Their arithmetic content is the same.

2.2 Fresh replay and practical assessment

Each supplied final machine was replayed on its own source. The symbolic checks quantify over arbitrary natural register values, a natural extra header gap, and a nonempty finite spare-register list; they are not bounded tests of small inputs. The reported inference counts are proof-construction work, not actual Turing-machine running-time comparisons.

Machine	States	Guarded contracts	Rows covered	Symbolic inferences
Contextual U211	211	5,084	422/422	1,277,525
Research U211	211	5,084	422/422	1,280,108
Rotated U211	211	5,084	422/422	1,245,944
New composed U209	209	5,084	418/418	1,250,049

Table 3: Fresh complete replays. Every minimum duration constant is 5.

The final source depth is 42 for Contextual and Research, 41 for Rotated, and 42 for U209; the maximum run of zero bits in an address is six in every case. Each blank initializer reaches the same calibrated frame after 48,140 literal steps. Full fresh receipts are included in `comparison/`; these counts should not be interpreted as a proof of optimality of any source association.

For a reader seeking the shortest local reduction to review, Rotated has the simplest last-state argument: a single zero arrival invariant and two three-step paths. Research offers a particularly explicit account

of the four-case selector fold and of why its all-tape theorem does not cover the preceding source rotations. Contextual is valuable both for the second selector pattern and for preserving the complete older calibration packet. All three contribute to the new construction; there is no need to choose one as the only useful predecessor.

The compact U212 archive included by Research and Rotated omits the full old U222 logical-calibration report and proof templates, while an embedded older report still says that archive is included. Contextual's nested ancestry supplies it. The present package preserves all three originals and additionally exposes the recovered original U222 ZIP directly, with its ancestry and hashes. This closes an inspection gap; it is not evidence that a supplied machine is mathematically incorrect.

2.3 Why the three saved states do not add automatically

Each proposal first changes the physical decoder graph. A native label therefore does not necessarily implement the same selector in another proposal. In particular, the Contextual and Research helpers occupy overlapping source regions, and retagging 218 can change a helper needed by another fold. Applying three final table patches without arranging their entry conditions and final helper paths is unjustified.

The useful commonality is more specific than equal state counts: all three savings exploit information already stored in the immediately preceding tape cell. The new problem is to produce one source layout that supplies all required predecessor bits while retaining the required helper behavior. That is the research step carried out next.

3 The new construction

3.1 Closed source factors and decoder rotations

The source is an ordered binary sequence tree. Leaves are the primitive operations `CI`, `CD`, `CB`, `CC`, and `CH`. Some control instructions name ancestors by counting left or right edges, so an arbitrary change of parentheses could change control flow. A factor is closed when no branch escapes its root and it contains no halt instruction. For three closed factors the reassociation

$$((A, B), C) \longleftrightarrow (A, (B, C))$$

preserves their ordered effects and resolved ordinal successors. Both guard outcomes of every decrement must be included.

A pure decoder reads a bit, preserves it, moves right, and enters the corresponding child. For $p = (q, c)$ and $q = (a, b)$, a left reassociation replaces these rows by $p = (a, q)$ and $q = (b, c)$. Its inverse is handled similarly. The inner state's complete physical incoming inventory must contain only the expected outer edge. Every occurrence of a shared outer decoder is changed in the source. The builder checks closure, pure rows, incoming edges, the ordered primitive word, and two independent source successor calculations after every rotation.

An intermediate table in this construction need not yet run its newly associated source correctly. In particular, the three Contextual retagging rotations must be treated as a group, followed by their entry repair. A prefix-decoding audit fails at an unfinished intermediate retagging stage; this prevents an incorrect per-stage runtime claim. The source control-flow identities remain valid at those stages. The complete prefix and whole-runtime checks apply to the finished tables. Source equality and runtime correctness are separate obligations.

3.2 A reproducible 30-rotation recipe

The common U212 native table has SHA-256

`386ce16259d4c97e4af31b3ba494feaff11a0a1a066e0235dddc154b4cea38c4.`

The portable builder starts from this exact table and its pinned source. It performs the following fixed operations, with no optimization search needed during reproduction.

1. Apply the nine Research rotations, obtaining its 212-state prefold layout. Do not delete 181 yet.
2. Reassociate the closed emitter region in six further rotations. Apply the label cycle

$$215 \mapsto 129 \mapsto 204 \mapsto 214 \mapsto 216 \mapsto 215.$$

   This is a literal table isomorphism, applied to source and target state labels; it changes no source address or behavior by itself.
3. Apply the three Contextual rotations $(206, 199, 1), (205, 206, 1), (7, 205, 1)$ and repair 171/0 from $0\,L\,7$ to $0\,L\,205$.
4. Transport the closed continuation rooted at 116 through twelve successive reassociations. Each uses $((A, B), C) \mapsto (A, (B, C))$; the same 78-primitive continuation is the factor C.
5. Apply all three guarded deletions simultaneously and densely rename the remaining 209 states.

The entire recipe, factor lengths, source occurrence prefixes, and stage hashes are in `machine/construction.json`. Appendix B prints the 30 rotation triples. The builder independently reaches exactly the table produced by the original direct construction.

3.3 The association that makes the composition possible

After the six new emitter rotations and label cycle, the relevant closed region has the following source decoder shape. Ordered pairs denote sequencing, and repeated factors retain their original implementations:

$$\begin{array}{lcllcl} 217 & = & (98, 129), & 129 & = & (158, 204), \\ 204 & = & (147, 214), & 214 & = & (146, 216), \\ 216 & = & (220, 215), & 215 & = & (163, 218), \\ 163 & = & (90, 211). & & & \end{array}$$

This layout simultaneously exposes the Contextual pair 163/180 and the Research pair 220/181 with opposite predecessor tags. The remaining obstacle is the one-tagged entry into 218. Moving continuation 116 down the enclosing right spine changes that entry to a zero-tagged one.

The transport also moves the Research helper. The finished local shape contains

$$217 = (158, 129), \quad 129 = (147, 204), \quad 216 = (163, 215), \quad 215 = (218, 116).$$

Consequently native **217**, rather than 129, now supplies the two-way selector needed by the Research fold: a predecessor zero leads through 158 to 218, while a predecessor one leads through 129 to 147. Recognizing this relocation is the decisive compatibility step. Keeping the old helper label would produce the wrong transition path.

Once 218 is replaced by 157, the Contextual zero case becomes a five-step path rather than a three-step path. This is an actual interaction between the folds. It is verified on the combined final table in the next section; the original isolated certificates are not merely concatenated.

The final source has the same 3,909 leaf occurrences and 197 distinct subtrees as its parent. Exactly 3,096 leaf addresses change. The saving comes from sharing physical control states using already written tape bits; it does not arise from deleting an arithmetic axiom or shortening the ordinal source program. Relative to the original U212 table, 38 retained rows change and six outgoing rows disappear.

3.4 The arithmetic-entry repair

The Contextual rotations include decoder 7, which has an additional left-moving runtime entry from 171/0. The repair targets 205 instead. For current zero and predecessor bit zero, the old path $171, 7, 86$ and new path $171, 205, 86$ have identical whole-tape endpoints. For predecessor one the corresponding paths are $171, 7, 205, 164$ and $171, 205, 206, 164$. They require respectively two and three transitions and read only the two specified cells. Both identities are checked against the actual baseline and final prefold tables. The full runtime check also rederives the affected primitives.

4 The simultaneous three-state fold

Let P be the released 212-state table after all source reassociations, and let C be the final 209-state table. Both start in native state 0; native 254 is halt. Define

$$f(180) = 163, \qquad f(181) = 220, \qquad f(218) = 157, \qquad f(q) = q \text{ otherwise.}$$

The map is deliberately not injective. Delete native $180, 181, 218$, redirect their incoming targets by f, and change only the two retained source rows

$$163/0 = (0, L, 211), \qquad 220/0 = (0, L, 217).$$

All other retained rows equal their old rows with their target renamed by f. In particular, both rows of donor 157 are unchanged.

The complete old incoming inventories establish the following arrival invariants, independently of the initial tape:

state	bit immediately left of head	incoming rows
163	0	216/0
180	1	132/1, 199/1
220	0	214/0
181	1	185/1
218	0	158/0, 211/0, 215/0, 220/0.

Every listed incoming row writes the displayed bit and moves right. State 0 belongs to none of these five classes, so their invariants hold vacuously initially and hold at every later visit in the old computation.

Relate configurations by identical head position and identical entire tape, with new state $f(q)$ when the old state is q. For each ordinary old state, one new step simulates one old step. At the five exceptional states, the two possible scanned bits give the ten paths below. Entries in the last column are the numbers of new transitions; each path ends at f of the old successor state, with exactly the old

successor head and whole tape.

q	read	new path	length
163	0	$163, 211, 157, 67, 151, 90$	5
163	1	$163, 211$	1
180	0	$163, 211, 184, 152$	3
180	1	$163, 211$	1
220	0	$220, 217, 158, 157$	3
220	1	$220, 176$	1
181	0	$220, 217, 129, 147$	3
181	1	$220, 176$	1
218	0	$157, 67, 151, 90$	3
218	1	$157, 148, 186, 86$	3.

Each nontrivial path reads only the original head cell and its immediate left neighbor; the incoming tag and the scanned bit exhaust all cases. Every write preserves the bit already present, so the unread exterior is arbitrary. No interior state is halt. For old 220/0, the successor is old 218 and the new path correctly stops at its image 157. The zero bit just left of that successor head is preserved, supplying the premise of the next 218 block.

Why ordinary donor entries are safe. Donor 157 already has ordinary old entries, including 65/1 and 209/0. No tag premise is imposed on these entries. If the old state is 157, the ordinary-row identity gives a one-step simulation because its two rows are unchanged. If the old state is 218, the new state is also 157, but the zero arrival tag gives the three-step block above. These are different cases of the simulation relation, distinguished by the old state and its actual run. The simulator need not reconstruct that old state from its current control label. Nor must a path stop whenever it passes a state that happens to be the image of some other old state: the synchronization point is fixed by the old transition being simulated.

Exact time change. For any finite old prefix of T transitions, let n_5 count occurrences of old state/read pair $(163, 0)$, and let n_3 count occurrences of

$$(180, 0),\ (220, 0),\ (181, 0),\ (218, 0),\ (218, 1).$$

Concatenating the blocks gives a synchronized new prefix of exactly

$$\tau(T) = T + 2n_3 + 4n_5, \qquad T \le \tau(T) \le 5T.$$

Thus, if T_{pre} is the old first halting time, the new first halting time is

$$T_{\mathrm{new}} = T_{\mathrm{pre}} + 2n_3 + 4n_5.$$

This is the time formula for the final simultaneous fold only. It makes no claim about the earlier source reassociations or the relative times of the three original U211 machines.

Finite reflection, including the internal PA claim. Forward simulation follows by induction on the number of old steps. For reflection, suppose the new machine halts by time s but the old machine has not halted by time s. Simulate those s old steps. Their blocks form a new prefix of length at least s with neither a halting synchronization point nor an interior halt, contradicting the proposed new

halt. This is a finite argument; an infinite-run or compactness assumption is unnecessary. It proves halting equivalence on every common initial binary tape from state 0.

For arithmetization, use any standard primitive-recursive encoding of finite-support tapes, head positions, states, and finite traces. The predecessor inventories and ten bounded paths are fixed finite facts, and the concatenation and reflection arguments use only natural-number induction. Alternatively, for a fixed horizon s, encode a finite input window containing all cells within distance $5s$ of the initial head; no simulated prefix can read outside that window. PA can quantify over such finite words and prove the bounded simulation statements uniformly. The external assertion for arbitrary infinite binary tapes follows from the same local reasoning, but it is not misrepresented as second-order quantification over infinite tapes inside PA. In particular, PA proves the required equivalence for the arithmetically specified all-zero tape.

The independent script reads only the two literal tables, derives the first synchronizing exceptional paths, checks every ordinary row and all incoming inventories, and verifies the delay classes. It imports neither the candidate builder nor its original checker. Its positive result is an exact finite audit supporting the proof above, not a proof-assistant kernel certificate.

5 Whole-tape primitive simulation and blank initialization

The local fold theorem compares the transformed 212-state table with U209. To connect the new source addresses to the original arithmetic program, an additional whole-runtime argument is required. This section describes the exact quantified contracts and their verification rules.

5.1 The quantified tape frame

For $v \in \mathbb{N}$ put $B(v) = 0\,1^{v+1}$ and $C(v) = 1\,0\,1^{v}$. For a finite list A, concatenate the corresponding blocks to obtain $B(A)$ and $C(A)$. At a primitive boundary, native control is 84 and the head points to the first bit of address p. The whole tape is

$$\cdots 0\,1\;p\;0^{56+g-|p|}\;1\;B(v_0)\cdots B(v_8)B(a_0)B(A)\;0\cdots .$$

Here $g, a_0, v_0, \ldots, v_8$ are arbitrary natural numbers and A is an arbitrary finite list, possibly empty. Thus the spare list (a_0, A) is nonempty. The logical registers x_i are arranged by

$$\pi = (6, 2, 0, 4, 3, 7, 8, 5, 1), \qquad v_{\pi(i)} = x_i.$$

For a decrement, the zero case and the positive case $x_i = u + 1$ are checked separately. Both zero exteriors are represented explicitly. The final maximum address length is 42 and the longest zero run is six, compatible with the inherited seven-zero origin detector and header 56.

The source checker compares every ordered effect and every ordinal control successor with the original U212 source. Its separate literal prefix audit checks all 3,908 internal tree prefixes on both the final prefold and U209 tables; the maximum such trace is 49 steps. It rejects a write that changes a prefix bit or a read into the unspecified exterior. This is a useful independent structural check, not a replacement for the full primitive contracts.

5.2 Soundness of the symbolic rules

Run lengths, displacement, and elapsed time are affine expressions in thirteen slots: a constant, gap, nine primary values, first-spare value, and the width of the remaining spare list. The width denotes that list's actual summed block length. The engine uses four elementary rules.

1. A literal transition reads an actual final-table row. A symbol can be removed from a run only when its affine length is provably at least one; an uncertain empty run or an opaque-context read is rejected.
2. A self-loop that moves in the traversal direction can consume an entire homogeneous run. Induction on its natural length proves exactly that many repetitions of the literal transition, including length zero.
3. Four generic one-block traversals are rederived from the final table for arbitrary v and opaque left/right contexts. They take $v + 2$ steps and return to their entry state. Concatenation and induction on a finite list justify traversing every spare-list length.
4. Word normalization joins adjacent equal runs, removes zero-length runs, absorbs finite zero runs into the zero exterior, and uses $C(A)1 = 1B(A)$ and its reversed identity. The latter follows by induction on A, beginning with the one-block identity.

For each nonhalting primitive, the symbolic endpoint must equal the expected complete tape, head displacement, and dispatcher state. Address overflow, positive decrement's gap change, increment's appended spare block, and control-ancestor cuts are included. A halting primitive must reach native 254. Every accepted duration has nonnegative coefficients and a strictly positive constant. Thus simulation cannot hide an infinite source computation in finitely many TM steps.

There are 3,909 primitive occurrences and 1,175 additional positive decrement cases, for **5,084 contracts**. The completed derivation uses **1,250,049 symbolic inference applications**, covers all **418 literal rows**, and has minimum duration constant **5**. The 4,000-inference limit per attempted contract bounds the proof search, not the natural register values represented by a successful derivation. The inherited assertion-based engine refuses Python optimization mode.

This is an executable induction argument with a small, explicitly described mathematical semantics. It is not a new trusted proof kernel. The final-table source/runtime check and the independent local fold check provide complementary evidence; transition coverage alone would not establish correctness.

5.3 The actual blank-tape entry

An exact sparse-tape replay of 48,140 transitions reaches native 84 with head -2, header coordinate 54, source ordinal 136, and address `001100110101010`. The logical store is

$$(1, 0, 13, 0, 0, 1, 0, 0, 1),$$

and the spare list consists of 43 values equal to 1. There are precisely 120 one-cells; the check compares their entire finite support against an independently reconstructed frame and fixes every exterior cell to zero. The original U212 and U209 prefixes agree, and no edited retained row occurs in this prefix.

The same independent audit replays 141 source steps from ordinal 106 and store $(1, 0, 4, 0, 0, 1, 0, 0, 1)$ to the above ordinal and store. This binds the literal initialization to the inherited safe-suffix argument. The startup is an interior command frame, not a clean root with all registers zero. Its nonzero cache must retain the specific interpretation used in the calibration; Section 6 explains why it is not already a proof of falsity.

Theorem 1 (Blank-start preservation)**.** *The final U209 table and the original calibrated U212 table have the same blank-tape halting behavior. The preservation argument is formalizable in PA.*

Proof. Their exact blank prefixes reach the same interpreted source occurrence and register/spare frame. The original and transformed source have the same ordered primitive effects and every resolved

successor. By induction on source execution, their register programs therefore agree at every ordinal boundary. The final whole-tape primitive contracts implement each such step with positive finite duration and preserve the frame. Consequently a source halt produces a literal halt; a literal halt cannot arise inside a nonhalting primitive contract. Finite trace induction gives preservation and reflection. All expressions involved are finite source/tape codes, natural lengths, and finite-list induction, so the argument is available in PA. The new table does not require an assumption that PA is consistent. □

6 Arithmetic calibration and its proof boundary

The state reduction and the arithmetic interpretation are separate parts of the argument. The reduction proved in this report preserves a particular literal proof searcher. It does not obtain a new arithmetic theorem merely by changing the name of the theory in the independently formalized K215 result. The following account identifies exactly what is inherited.

6.1 The unary theory in the question

Work in classical equality-free first-order logic over a nonempty domain, with one binary relation E. Write $x \in a$ for $E(x, a)$ and

$$x \subseteq a \quad \Longleftrightarrow \quad \forall z\,(z \in x \longrightarrow z \in a).$$

For every formula $\varphi(x)$ whose only possible free variable is x, the theory U has the axiom

$$\forall a\,\exists b\,[\forall x\,(x \in b \longrightarrow \varphi(x)) \ \wedge \ \forall x\,(x \subseteq a \longrightarrow (\varphi(x) \longrightarrow x \in b))]\,.$$

Bound names are freshened. In particular, the source a is not a free matrix parameter. The output may contain further objects satisfying φ; the schema is weak covering, not exact separation.

The supplied unary manuscript [4] gives a conventional proof of

$$\mathrm{PA} \vdash \mathrm{Con}(\mathsf{U}) \longleftrightarrow \mathrm{Con}(\mathrm{PA}).$$

Its lower interpretation has a dependency order that is essential to its validity. Coextensionality means equality of member profiles. Numerical representatives are saturated, transitive objects whose members are linearly ordered by membership modulo coextensionality, and whose intersections with represented initial segments have endpoints. Saturation, rather than an assumed global extensionality principle, supplies the congruences used in the argument.

Local predecessor facts come first. For a unary predicate Q, a cover of its failures below an alleged counterexample gives a least failing member if there is one. If there is no failing member, the counterexample itself has a predecessor satisfying Q. The two cases prove induction without assuming that every number already has a successor. Parameters are then internalized by the unary formula

$$Q(v) = \forall \bar{p}\,(A_\theta(\bar{p}) \longrightarrow \theta(v, \bar{p})),$$

where A_θ is the conjunction of the base and step premises. Thus this argument does not illegally insert free parameters into a matrix of U.

Induction yields comparison and a least numerical tuple in every definable, invariant nonempty class of such tuples. Suppose an exact bounded numerical construction fails, and let $B(\bar{n})$ express that failure. A legal unary matrix is

$$\exists \bar{n}\,(\bar{n} \text{ is a least } B\text{-tuple} \ \wedge \ \chi(z, \bar{n})).$$

All least tuples are coordinatewise coextensional. Invariance identifies their selected profiles, and a supporting source makes the weak cover exact for that profile, contradicting failure. The supporting source is chosen in the proof; it is not a free variable of this matrix. This derived comprehension supplies successors, numerical powers, saturated pair codes, and graphs. Canonical finite histories quantify over earlier graph witnesses instead of placing an arbitrary graph among the forbidden free matrix parameters. Iteration then supplies addition and multiplication with their usual recursion equations. The induction already proved supplies every translated PA induction instance.

In the opposite direction, PA interprets $x \in a$ by binary-bit membership. It proves that $x \subseteq a$ implies $x \leq a$. For each fixed translated matrix, induction constructs a bitset selecting precisely the desired indices at most a. The induction instance depends on that matrix; no uniform truth predicate is asserted. The syntax transformations in both directions use fixed formulas, capture-avoiding renaming, substitution, and finite proof concatenation. Their correctness is justified by structural and proof-length induction available in PA. This is an ordinary proof of internal consistency equivalence, rather than only an argument about standard models. No implementation of the complete arbitrary-proof translator or new proof-assistant certificate is supplied.

6.2 The theory actually searched by the literal machines

The three U211 proposals [1, 2, 3] use a common numerical calculus $\mathcal{D}_U$, not the separate positive dyadic stream described later in the unary manuscript. The numerical calculus uses truth code 0, falsity target 1, and Cantor pairing

$$P(x, y) = \frac{(x + y)(x + y + 1)}{2} + y.$$

Implication, membership, universal quantification, and auxiliary equality have respective constructors $4P(x, y)$, $4P(x, y) + 1$, $4P(x, y) + 2$, and $4P(x, y) + 3$. The alias $I(0, 0) = 0$ is intentional. Its handling is part of the numerical logical bridge, not a reason to treat the integer zero as an ordinary injective formula constructor.

Let W allow arbitrary additional matrix parameters, including the source, in the weak-cover schema, and let $\mathsf{IR} = \forall x \neg E(x, x)$. The machine-facing calibration is

$$\mathrm{Con}(\mathrm{PA}) \longrightarrow \mathrm{Con}(\mathsf{W} + \mathsf{IR}) \longrightarrow \mathrm{Con}(\mathsf{U}) \longrightarrow \mathrm{Con}(\mathrm{PA}).$$

The first implication is the bit-membership interpretation, which also validates irreflexivity and equality-containing matrices. The second is inclusion of the unary schema, and the third is the lower interpretation just described. Each implication has the finite-refutation transport needed to prove this cycle inside PA.

The retained native-calculus bridge additionally proves that a numerical proof of 1 is equivalent to a refutation of this machine-facing theory. It handles small-code aliases, dominated propositional axioms, recovery of membership vacuity, and equality through two-sided membership indiscernibility. Code 1 is interpreted as diagonal membership and becomes false under IR; conversely, numerical identity at 1 quotes a diagonal negation and generalization recovers irreflexivity. The U214 role reassignment also handles the permitted coincidence of source and output labels: its captured subset guard becomes emptiness under irreflexivity, and an ordinary cover at an empty source proves the exceptional instance. These facts are substantive parts of the calibration; none follows merely from a count of states or transitions.

6.3 Startup, soundness, and fair enumeration

The inherited blank initializer reaches an interior axiom-command frame, not a clean proof-search root. In particular, the incoming cache value 1 is an arithmetic selector and cannot be declared a proved contradiction. The retained source-suffix proof consumes its selector fields and builds nine explicit theorem heads above numerical tail 4. That tail is unsafe if exposed, since $4 = P(1, 1)$.

The scheduler theorem maintains the length of an explicit wrapper list above this opaque tail. Conservative command weights are -1 for selection, 0 for modus ponens, and $+10$ for a completed ordinary axiom command. A joint induction on Cantor diagonals bounds the debt from candidate 1 by two, so the nine-head seed leaves margin seven. The wrapper length is a ghost variable: it need not be recovered from the numerical stack, since $P(0, 0) = 0$ allows encodings with different apparent list lengths. Exact unpairing, not injectivity of list-length encoding, supports the invariant.

Soundness follows because the top, cache, and all explicit stack heads remain theorem codes and only cache value 1 can cause halting. Completeness uses a stronger compiler contract: a proof of a theorem can be compiled above every incoming numerical top and stack after counter and cache have been reset. A refutation therefore supplies cofinally many successful candidate indices. Every unsuccessful candidate terminates by descent of its continuation code, so the scheduler reaches a successful candidate beyond every finite initial cutoff. Preserving only axiom generation, without these startup and compiler facts, would not suffice.

6.4 What the new candidate proves and what remains inherited

For the final candidate M, the proof must separately establish its literal table, the source-preservation statement, every required whole-tape primitive simulation contract, and the complete blank-start frame. With those new obligations proved, the calibrated conclusion is the composition

$$\begin{aligned} \mathrm{Halt}(M, \mathrm{blank}) &\longleftrightarrow \text{the retained source continuation halts} \\ &\longleftrightarrow \exists p \ \mathrm{Prf}_{\mathcal{D}_U}(p, 1) \\ &\longleftrightarrow \neg\, \mathrm{Con}(\mathsf{W} + \mathsf{IR}) \\ &\longleftrightarrow \neg\, \mathrm{Con}(\mathrm{PA}). \end{aligned}$$

The local transition lemmas have a small finite proof obligation. The unbounded runtime result uses the symbolic interpreter's natural-number and list induction rules. The arithmetic endpoint additionally uses the inherited theory, codec, source-compiler, and protected-tail proofs. These are different verification layers. Successful Python checks are not an exported PA derivation or a proof-assistant kernel certificate. The present conventional mathematical argument is PA-formalizable at this explicitly stated boundary; it neither assumes nor proves that PA is consistent.

The full U222 packet is retained to make the oldest numerical-calculus dependency inspectable. This matters because the compact combined U212 packet preserves only three U222 source/data files, while some old reports embedded in it still refer to an included complete U222 archive. Adding the original U222 packet closes that documentation omission without changing any machine or mathematical premise.

7 Reproduction, provenance, and limitations

7.1 One-command replay

The delivered package needs Python 3.10 or later and its standard library for verification. From the extracted package directory run

```
python3 reproduce.py
```

The script first verifies frozen file hashes, rebuilds the table from the common U212 inputs, checks source preservation and final literal prefixes, checks the local fold (also under `-O`), runs the independently written fold auditor, replays blank startup, derives every symbolic primitive contract, and verifies full final-row coverage. It performs no network requests and no optimization search. The runtime replay takes a few minutes in the preparation environment; machine performance varies.

For an explicitly limited check use

```
python3 reproduce.py --quick
```

Its receipt says that runtime replay was skipped. It is not a substitute for the full command. The full command rejects `-O` and `PYTHONOPTIMIZE`, because the inherited symbolic/source code uses checked assertions. The small finite fold checkers use explicit checks and retain their checks under optimization.

To rebuild the PDF, install a normal LaTeX distribution and run

```
python3 reproduce.py --quick --pdf
```

The editable report is `report.tex`, with two proof sections and generated exact tables under `paper/`. The generated PDF and replay evidence are excluded from the frozen hash list because compilation metadata and timing fields can change. All frozen source, input, machine, and retained reference files are covered. The runtime receipt binds the exact final table and source by SHA-256.

7.2 Contents and attribution

Directory or file	Contents
`machine/`	Complete native/dense U209 tables, final 212-state prefold table, source sites, state map, 30-rotation construction ledger.
`inputs/`	Exact common U212 table and source; no repository checkout is needed.
`code/`	Portable builder, independent source and local-fold auditors, startup audit, retained native symbolic engine, certificate summary checker.
`evidence/`	Fresh exact and symbolic receipts, per-contract runtime data, independent mathematical audit and local proof notes.
`comparison/`	Three-proposal census, exact fingerprints and fresh baseline runtime/startup receipts.
`reference/`	All four originally requested archives, the recovered complete U222 calibration packet, commit-pinned manifest and retained license/attribution notices.
`SHA256SUMS`	Frozen identities checked before rebuilding.

The inherited theory U, numerical proof search, compilers, symbolic engine, and individual U211 reductions are credited to the supplied packets and their retained upstream notices. These include the ProveIt research lineage, Mira's consistency-machine architecture, Elliot Glazer's compact-theory work, Jeff's busy-beaver verification lineage, Andrew James Wade's compiler ancestry, BusyCoq, and the Coq Library of Undecidability [8, 7, 1, 2, 3]. The new work here is the compatible association, continuation transport, relocated selector, simultaneous fold proof and integration/audit. Existing licenses are preserved; no broader relicensing of inherited material is asserted.

7.3 Exact scope of the conclusion

The reduction meets the requested state-count target with a literal 209-state candidate. At the ordinary mathematical proof level, the new preservation results compose with the supplied finite-refutation translations to establish

$$\mathrm{PA} \vdash \neg\,\mathrm{Halt}(U209, \mathrm{blank}) \longleftrightarrow \mathrm{Con}(\mathsf{U}) \longleftrightarrow \mathrm{Con}(\mathrm{PA}).$$

This claim depends on the retained U-specific calibration, whose detailed audit is included. The finite and symbolic scripts do not recheck that entire logical chain in a proof-assistant kernel. The result is therefore presented as a documented research candidate with conventional proofs and reproducible certificates, with the remaining formal-verification boundary stated explicitly.

No minimum-state theorem follows. Our unsuccessful direct-alias searches only ruled out their tested local templates, and are not treated as lower bounds. A final-fold slowdown of at most five applies only against the transformed 212-state intermediate. The source reassociations change addresses, so it would be incorrect to apply that bound to the original U212 or U211 computations.

8 Further research questions

1. **Can this 209-state table receive a small kernel certificate?** The ten-case local fold and the two-cell entry repair are compact initial targets. A reusable theorem about noninjective state maps with bounded whole-tape simulation blocks would make later reductions inexpensive to formalize. The broader task is the inherited numerical calculus and safe-start calibration, not just checking 418 rows.
2. **Can source reassociation search preserve helper behavior explicitly?** The successful composition needed the helper to move from 129 to 217. A search state should record demanded local paths and arrival tags as constraints, instead of freezing donor labels. This could find other compatible folds while making failed proposals easier to explain.
3. **Can two-cell certificates be extended to a small regular language of predecessor contexts?** The present invariants use one previously written bit. Longer source suffixes may supply more context, but any resulting reduction must prove that every actual incoming runtime edge satisfies the context language. Bounded execution samples would not establish that invariant.
4. **Can the runtime and source proof be packaged as one checkable certificate?** The current symbolic engine has four simple rules. Exporting their derivation traces to a small independently verified checker would reduce trust in Python and make the exact unlimited quantification easier to audit.
5. **Does a different calibrated arithmetic source reduce the 197-subtree bottleneck?** This work leaves all 3,909 primitives unchanged. A theory- or proof-calculus-level change remains

a separate opportunity, but it would require fresh soundness, completeness, nonzero-start, and fair-search proofs. A shorter axiom schema alone does not predict the number of physical Turing-machine states.

References

[1] *U211 ConPA Contextual*, supplied archive, `docs/incoming/U211_ConPA_Contextual_2026-10-28.zip`, ProveIt snapshot `da4ab9dd68cc`. Exact archive retained in `reference/`.

[2] *U211 ConPA Research / Decoder Reassociation*, `docs/incoming/U211_ConPA_Research_2026-10-28.zip`, same pinned snapshot. Exact archive retained in `reference/`.

[3] *U211 ConPA Rotated*, `docs/incoming/U211_ConPA_Rotated_2026-10-28.zip`, same pinned snapshot. Exact archive retained in `reference/`.

[4] *Unary power-cover theory*, `docs/incoming/pa_unary_power_cover.zip`, same pinned snapshot. The complete manuscript and proof material are retained in `reference/`.

[5] Elliot Glazer, *K215 halts iff ZFC is inconsistent*, Metamath Zero/PA specification and checking receipt. https://github.com/elliotglazer/k215-pa.

[6] ProveIt, *K215 verification*. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Computability/BusyBeaver/Research/ZFC/k225-formal/verification/k215.

[7] Elliot Glazer, *K230*, especially "The theory K". https://github.com/elliotglazer/k230-rocq.

[8] Jeff's busy-beaver project, *ZF/m254*, branch `zf-254`. https://github.com/jeff87654/busy-beaver/tree/zf-254/ZF/m254.

A Complete dense transition table

An action $w\,D\,q$ writes w, moves in direction D, and enters q. The working labels are 0 through 208; label 209 is halt and has no outgoing rows. Initial state, head coordinate, and blank symbol are all 0.

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
0	$1\,R\,92$	$1\,R\,0$	70	$0\,L\,89$	$1\,R\,78$	140	$0\,R\,96$	$1\,R\,159$
1	$0\,L\,1$	$1\,L\,2$	71	$0\,R\,55$	$1\,L\,79$	141	$0\,R\,175$	$1\,R\,166$
2	$0\,L\,3$	$1\,L\,2$	72	$0\,R\,80$	$1\,R\,81$	142	$0\,R\,12$	$1\,R\,159$
3	$0\,L\,5$	$1\,L\,2$	73	$1\,R\,0$	$1\,R\,63$	143	$0\,R\,88$	$1\,L\,166$
4	$1\,R\,6$	$0\,L\,4$	74	$0\,R\,74$	$1\,R\,82$	144	$0\,R\,88$	$1\,R\,158$
5	$0\,L\,9$	$1\,L\,2$	75	$0\,L\,145$	$1\,L\,141$	145	$0\,L\,152$	$1\,L\,163$
6	$0\,R\,6$	$0\,R\,10$	76	$0\,R\,85$	$1\,R\,86$	146	$0\,R\,84$	$1\,R\,88$
7	$0\,R\,190$	$1\,R\,159$	77	$0\,R\,20$	$1\,R\,38$	147	$0\,R\,159$	$1\,R\,12$
8	$0\,R\,14$	$1\,R\,13$	78	$0\,R\,89$	$1\,L\,132$	148	$0\,R\,83$	$1\,R\,143$
9	$0\,L\,15$	$1\,L\,2$	79	$0\,L\,108$	$1\,R\,56$	149	$0\,R\,144$	$1\,R\,155$
10	$0\,L\,16$	$0\,R\,10$	80	$0\,R\,49$	$0\,R\,33$	150	$0\,L\,66$	$1\,L\,141$
11	$0\,L\,26$	$1\,R\,128$	81	$0\,R\,91$	$0\,L\,42$	151	$0\,R\,150$	$1\,R\,170$
12	$0\,R\,19$	$1\,R\,20$	82	$0\,R\,178$	$1\,R\,77$	152	$0\,R\,159$	$1\,R\,114$
13	$0\,L\,89$	$1\,L\,31$	83	$0\,R\,96$	$1\,R\,12$	153	$0\,R\,97$	$1\,R\,97$
14	$0\,L\,11$	$0\,R\,22$	84	$0\,R\,88$	$1\,R\,97$	154	$0\,R\,49$	$0\,R\,32$
15	$0\,L\,23$	$1\,L\,2$	85	$0\,R\,207$	$1\,R\,21$	155	$0\,R\,192$	$1\,R\,188$
16	$1\,L\,1$	$0\,L\,1$	86	$0\,R\,116$	$1\,R\,100$	156	$0\,L\,195$	$1\,R\,195$

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
17	0 R 24	0 R 32	87	0 R 174	1 R 167	157	0 L 187	1 R 96
18	0 R 98	1 R 113	88	0 R 102	1 R 103	158	0 R 168	1 R 179
19	0 R 28	1 R 29	89	0 R 80	1 R 79	159	0 R 19	1 R 171
20	0 R 30	1 R 31	90	0 R 101	1 R 196	160	0 R 164	1 R 187
21	0 R 24	0 R 59	91	0 R 105	1 R 106	161	0 R 159	1 R 96
22	0 R 33	1 R 22	92	0 R 65	0 R 0	162	0 R 125	1 R 164
23	0 L 34	1 L 2	93	0 R 27	1 R 107	163	0 R 97	1 R 162
24	0 L 35	0 R 0	94	0 R 183	0 L 208	164	0 L 190	1 R 142
25	0 R 36	1 R 25	95	1 L 109	1 L 95	165	0 L 66	1 R 173
26	0 R 37	1 R 38	96	0 R 19	1 R 110	166	0 R 146	1 R 172
27	0 R 39	0 L 40	97	0 R 60	1 R 60	167	0 R 96	1 R 201
28	0 R 41	1 R 42	98	0 R 68	1 R 135	168	0 R 161	1 R 143
29	0 R 43	1 R 44	99	0 R 93	1 R 108	169	0 R 148	1 R 87
30	0 R 45	1 R 46	100	0 R 105	0 L 208	170	0 R 161	1 R 96
31	0 R 21	1 R 47	101	0 R 117	1 R 118	171	0 R 181	1 R 122
32	0 R 48	1 R 32	102	0 R 71	1 R 58	172	0 L 193	1 R 60
33	0 R 32	1 R 33	103	0 R 57	1 R 119	173	0 R 145	1 R 88
34	0 L 74	1 L 2	104	0 R 21	1 R 120	174	0 R 152	1 R 202
35	1 R 35	1 L 1	105	0 R 21	0 L 10	175	0 R 140	1 R 84
36	0 R 50	1 R 36	106	0 R 61	1 R 42	176	0 R 175	1 R 75
37	0 R 17	1 R 111	107	0 R 121	1 R 8	177	0 R 126	1 R 169
38	0 R 13	1 R 52	108	0 R 101	1 R 122	178	0 R 138	1 R 116
39	0 L 11	0 R 36	109	0 L 208	0 L 95	179	0 R 97	1 R 51
40	0 L 4	1 L 53	110	0 R 124	1 R 104	180	0 R 168	1 R 51
41	0 R 24	0 R 25	111	0 R 115	1 L 72	181	0 R 17	1 L 131
42	0 R 13	0 L 16	112	0 R 165	1 R 181	182	0 R 143	1 R 149
43	0 R 55	1 R 56	113	0 R 160	1 R 184	183	0 R 76	1 R 116
44	0 R 57	1 R 58	114	0 R 148	1 R 146	184	0 R 106	1 R 124
45	0 R 24	0 R 54	115	0 R 11	1 R 129	185	1 L 40	1 L 109
46	0 R 60	1 R 61	116	0 L 131	0 L 1	186	0 R 175	1 R 156
47	0 R 62	0 L 16	117	0 R 39	1 R 72	187	0 R 182	1 R 183
48	0 R 25	1 R 48	118	0 R 61	1 R 132	188	0 R 66	1 R 205
49	0 L 35	0 R 63	119	0 R 61	1 R 133	189	0 R 139	1 R 198
50	0 R 16	1 R 50	120	0 R 70	0 L 16	190	0 R 84	1 R 191
51	0 R 147	1 R 143	121	0 R 14	1 R 132	191	0 R 157	1 R 186
52	0 R 67	1 R 68	122	0 R 21	1 R 134	192	0 R 64	1 R 7
53	0 L 40	1 L 53	123	0 R 140	1 R 189	193	0 R 150	1 R 127
54	0 R 22	1 R 54	124	0 R 137	1 L 20	194	0 R 18	1 R 94
55	0 L 11	0 R 50	125	0 R 130	1 R 180	195	0 R 150	1 R 173
56	0 R 60	0 L 16	126	0 R 204	1 R 156	196	0 R 104	1 R 96
57	0 R 69	1 R 70	127	0 R 75	1 R 176	197	0 R 90	1 R 88
58	0 R 61	1 R 71	128	0 R 75	1 R 206	198	0 R 202	1 R 200
59	0 R 54	1 R 59	129	0 R 203	1 R 157	199	0 R 150	1 R 112
60	0 R 49	0 R 25	130	0 R 12	1 R 141	200	0 R 156	1 R 199
61	0 L 28	1 L 72	131	0 R 30	1 R 69	201	0 R 151	1 R 123
62	0 L 117	1 L 30	132	0 R 39	1 L 124	202	0 L 201	1 R 168
63	0 R 73	1 R 63	133	0 R 153	0 L 16	203	0 R 197	1 R 28
64	0 R 97	1 R 150	134	0 R 154	0 L 16	204	0 R 163	1 R 193
65	1 R 185	1 L 53	135	0 R 88	1 R 136	205	0 R 146	1 R 177
66	0 R 144	1 R 145	136	0 R 197	1 R 140	206	0 R 161	1 R 194
67	0 R 76	1 R 209	137	0 R 24	0 R 48	207	0 R 45	1 R 100
68	0 L 197	1 R 31	138	0 R 99	1 R 37	208	0 L 4	0 L 208
69	0 R 41	1 R 62	139	0 R 172	1 R 148			

B Exact rotation recipe

Each triple is outer state, inner state, and old edge side. Side 0 means $((A, B), C) \mapsto (A, (B, C))$; side 1 means its inverse. Apply the displayed five-cycle after step 15, and the 171/0 repair after step 18. The local three-state fold follows step 30. Only source closure/CFG preservation is claimed at every unfinished stage; full runtime is checked for the final table.

Step	Phase	Outer	Inner	Side	Factor lengths
1	Research	215	204	1	160, 163, 159
2	Research	217	181	1	52, 80, 160
3	Research	185	217	0	132, 160, 482
4	Research	174	185	1	162, 132, 642
5	Research	129	158	0	26, 56, 78
6	Research	217	215	1	160, 323, 159
7	Research	204	214	0	52, 108, 163
8	Research	215	129	0	26, 134, 323
9	Research	216	220	1	56, 80, 27
10	New emitter association	217	215	0	26, 457, 159
11	New emitter association	215	129	0	134, 323, 159
12	New emitter association	129	204	0	52, 271, 159
13	New emitter association	204	214	0	108, 163, 159
14	New emitter association	214	216	0	136, 27, 159
15	New emitter association	216	163	1	27, 103, 56
16	Contextual tags	206	199	1	81, 123, 26
17	Contextual tags	205	206	1	80, 204, 26
18	Contextual tags	7	205	1	29, 284, 26
19	Continuation transport	162	203	0	503, 2515, 78
20	Continuation transport	203	223	0	401, 2114, 78
21	Continuation transport	223	188	0	56, 2058, 78
22	Continuation transport	188	177	0	1042, 1016, 78
23	Continuation transport	177	89	0	80, 936, 78
24	Continuation transport	89	174	0	294, 642, 78
25	Continuation transport	174	217	0	26, 616, 78
26	Continuation transport	217	129	0	134, 482, 78
27	Continuation transport	129	204	0	52, 430, 78
28	Continuation transport	204	214	0	108, 322, 78
29	Continuation transport	214	216	0	136, 186, 78
30	Continuation transport	216	215	0	130, 56, 78